Underdetermination of Finite Sequences and Construct Validity in Intelligence Testing: an Accurate and Accessible Discussion of Interpolation, the g Factor, and Numerical Completion Items

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Abstract

Numerical sequence completion items are among the most familiar tasks in reasoning tests: a respondent is shown a finite set of terms and asked to state the next one. This article develops a precise claim, formulated in a way that remains accessible while preserving mathematical rigor: *given only the observed terms, the next term is not uniquely determined as a matter of logic unless additional assumptions are imposed on the class of admissible rules or on the criterion used to select a rule.* The argument is established through Lagrange interpolation and the explicit construction of infinite families of functions compatible with the same data.

The second part of the paper addresses the psychometric question. It explains what the g factor is, why it was introduced in the intelligence literature, how it is estimated through factor analysis, and in what sense it differs both from a simple IQ score and from an "essence" directly observable in a person. Conceptually, the central point is the following: the psychometric usefulness of an item does not automatically eliminate the problem of how that item should be interpreted. An item may correlate with other tests, contribute to the estimate of g, and even have predictive value, while still being conceptually narrower than ordinary language may suggest.

The conclusion is cautious but firm. Sequence items may measure real and useful abilities, such as inferring the rule *intended* by the test designer, rapidly recognizing conventional regularities, and reasoning under time constraints. However, without explicit constraints on the space of admissible solutions, such items do not by themselves justify the claim that one specific continuation is the only mathematically correct answer. For this reason, their interpretation as a direct measure of "general intelligence" requires more theoretical care than is often made explicit.

Keywords: Lagrange interpolation, numerical sequences, underdetermination, g factor, factor analysis, construct validity, intelligence testing.

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1. Introduction

Items of the form *"complete the sequence"* seem simple, almost obvious. A sequence such as "2, 4, 6, 8, ..." strongly suggests that there is only one possible continuation and that the respondent's task is simply to identify it. This impression is so strong that, in many educational and psychometric settings, it becomes an implicit part of the very definition of a "correct" answer.

Yet the issue is more delicate. In mathematics, a finite number of data points does not by itself determine a unique law unless one also specifies the family of admissible models and a criterion for preferring one compatible model over another. In other words, the question "what is the next term?" almost always contains hidden assumptions. If those assumptions are not made explicit, the demand for a unique answer is not a purely logical fact but a convention.

This observation has two distinct consequences, which should be kept separate. The first is mathematical: we must clarify what genuinely follows from the observed data. The second is psychometric: we must ask what exactly is being measured by a test that awards full credit to someone who identifies the expected convention and zero credit to someone who questions the uniqueness of the problem.

This article has three aims:

1. to show, formally and clearly, why the continuation of a finite sequence is underdetermined unless further constraints are added;
2. to explain precisely what the g factor is, where it comes from, and how it is estimated;
3. to discuss, in a balanced way, what this implies for the construct validity of numerical completion items.

The paper is written for an educated reader who may not be a specialist. For that reason, each technical step is accompanied by an intuitive explanation, and the limits of the argument are made explicit rather than left implicit.

2. Related Work

The literature relevant to this paper belongs to several different families: public or essayistic critiques of IQ testing, theoretical critiques of the reification of the g factor, and broader psychological models of intelligence. It is useful to distinguish them, because not all of them directly address the mathematical problem of underdetermination.

Public and non-formalized critiques of non-uniqueness. Taleb [13] raised, in a polemical form, the example of the sequence $\{1, 2, 3, 4, x\}$ as a case of a question with no unique answer. The intuition is close to the one developed here, but his intervention remains outside the formal academic literature and does not provide a mathematical formalization of the problem.

Critiques of the reification of the g factor. Gould [14] argued that the g factor is often treated as if it were a natural entity directly measured by tests, whereas it depends on statistical procedures and theoretical choices. The present paper is more limited: it does not claim that factor analysis is devoid of value, but rather that the mere fact that an item loads on g does not settle the question of its cognitive meaning. In that sense, the discussion of construct validity in Section 8 is close in spirit to Gould's critique without adopting its strongest form.

Broader models of intelligence. Sternberg [6, 7] and Gardner [15] argued, in different ways, that conventional tests capture only part of the abilities relevant to intelligence. These works do not directly treat the underdetermination of sequence items, but they are relevant as theoretical background: they show that the equation between high performance on standardized tasks and "intelligence" in a fully general sense has long been contested.

Informal didactic and mathematical observations. Knill [16] noted, in a teaching context, that many numerical items can be solved mechanically with tools such as finite differences, which weakens their value as transparent indicators of intelligence. Here too, however, one does not find a general treatment of the non-uniqueness of continuation.

Specific contribution of the present work. Relative to these lines of criticism, the present contribution is narrower but also more formal: to show that, in the absence of explicit constraints on the class of admissible rules, sequence-completion items are underdetermined. Proposition 1 and the discussion that follows provide the mathematical core

of the result, while the computational visualization gives an intuitive picture of its scope. Rather than a "total refutation" of psychometrics, the contribution is therefore a formal clarification of the hidden assumptions built into these items.

3. The Problem in Plain Language

3.1. Why these questions seem to have only one answer

When we look at a short sequence, we spontaneously search for the most economical rule that explains it. If we read

$$1,\ 3,\ 6,\ 10,\ \ldots$$

many people immediately recognize the triangular numbers and answer 15. That answer is reasonable, elegant, and in many practical contexts very likely the one *intended* by the author of the item.

The key point, however, is that "reasonable" does not mean "logically forced by the data". The observed sequence is compatible with the triangular-number rule, but not only with that rule. It is compatible with infinitely many alternative rules, some simple, some complicated, some artificial, others perfectly legitimate from a mathematical point of view.

3.2. What "underdetermined" means

Definition 1 (Underdetermination)**.** A problem is *underdetermined* when the available data are insufficient to identify a unique solution without further assumptions.

In the case of finite sequences, the data are the observed terms. The additional assumptions may take many forms:

- "we allow only arithmetic progressions";
- "we allow only polynomials of the lowest possible degree";
- "we choose the simplest rule according to a parsimony criterion";
- "we assume that the test designer had in mind the most conventional pattern".

These assumptions may be useful. But they are not contained in the observed terms themselves: they are added *beyond* the data.

3.3. The decisive distinction

This yields a distinction that will guide the rest of the paper:

> **Guessing the rule intended by the test** is not the same thing as **proving that the rule is the only logically possible one**.

Psychometric tests often evaluate the first ability in practice. Our analysis shows that it should not automatically be confused with the second.

4. Mathematical Foundation: Why a Finite Sequence Does Not by Itself Determine the Next Term

4.1. From a sequence to points in the plane

A finite sequence

$$a_0, a_1, \ldots, a_{n-1}$$

can be represented as the set of points

$$(0, a_0), (1, a_1), \ldots, (n-1, a_{n-1}).$$

The question "what is the next term?" then becomes:

> what value should a function take at the point $x = n$, given that at the points $0, 1, \ldots, n-1$ it passes exactly through the observed values?

Stated in this way, the issue becomes more transparent: we are asking a function to match a set of observed data and then predict a value outside the observed sample.

4.2. The classical Lagrange interpolation theorem

Let a set of distinct points be given such that

$$(x_1, y_1), \ldots, (x_n, y_n) \in \mathbb{R}^2, \qquad x_i \neq x_j \quad \text{whenever } i \neq j.$$

There exists a unique polynomial of degree at most $n-1$ that passes through all of them. Its explicit form is

$$P(x) = \sum_{i=1}^{n} y_i \prod_{\substack{j=1 \\ j \neq i}}^{n} \frac{x - x_j}{x_i - x_j}. \tag{1}$$

Theorem 1 (Uniqueness of the minimum-degree interpolating polynomial)**.** *There exists a unique polynomial P of degree $\leq n-1$ such that $P(x_i) = y_i$ for every $i = 1, \ldots, n$.*

This theorem is extremely important, but it is also frequently misunderstood. It does not say that there is only one function compatible with the data. It says something narrower and more precise: *within the class of polynomials of degree at most $n-1$, there is exactly one.*

4.3. The infinite family of functions that agree on the observed points

Define the nodal polynomial

$$W(x) = \prod_{i=1}^{n} (x - x_i). \tag{2}$$

By construction, $W(x_i) = 0$ at every observed node. Therefore, if we take the interpolating polynomial $P(x)$ and add to it any perturbation of the form $H(x)W(x)$, where H is any function well defined at the points of interest, we obtain another function that still passes through all the observed points.

In the simplest case, we choose $H(x) = k$ constant and obtain the family

$$P_k(x) = P(x) + k\,W(x). \tag{3}$$

Since $W(x_i) = 0$, it follows immediately that

$$P_k(x_i) = P(x_i) + kW(x_i) = y_i \quad \text{for every } i.$$

Proposition 1. *For every real value y^* and every point x^* distinct from the observed points, there exists a polynomial in the family* (3) *such that*

$$P_k(x^*) = y^*.$$

Proof. It is enough to choose

$$k = \frac{y^* - P(x^*)}{W(x^*)}.$$

Because x^* is not one of the observed nodes, we have $W(x^*) \neq 0$, so this value of k is well defined. □

Corollary 1. *Without further constraints on the class of admissible functions, the observed data do not uniquely determine the next term of a finite sequence.*

4.4. What this result really implies

The previous corollary must be interpreted carefully. It does not mean that, in ordinary language, all continuations are equally sensible. Nor does it mean that every test author would accept every answer. It means something more precise:

> the choice of a unique continuation does not follow from the observed data *alone*; it follows from the observed data *plus* an additional rule for selecting among compatible models.

That extra rule may be a parsimony principle, an appeal to culturally familiar patterns, or the intention of the item writer. But in every case it is additional information.

4.5. A completely explicit example

Consider the sequence

$$1,\ 3,\ 6,\ 10.$$

If we place the observed points at $x = 1, 2, 3, 4$, the minimum-degree polynomial that interpolates them is

$$P(x) = \frac{x(x+1)}{2},$$

that is, the familiar formula for triangular numbers. In this model,

$$P(5) = 15.$$

But the polynomial

$$W(x) = (x-1)(x-2)(x-3)(x-4)$$

vanishes at all observed points. Therefore, for every real number b, the function

$$Q_b(x) = \frac{x(x+1)}{2} + \frac{b-15}{24}(x-1)(x-2)(x-3)(x-4) \tag{4}$$

has the following properties:

1. it agrees with the observed data at the first four points;
2. it takes the value $Q_b(5) = b$.

If we choose $b = 15$, we recover the conventional continuation. If we choose $b = 42$, we obtain another function that is perfectly compatible with

the first four terms but yields a different fifth term. This example makes the core of the argument visible without ambiguity.

4.6. Why a simplicity criterion does not solve the logical problem

A natural objection is: *"fine, then let us simply choose the simplest solution"*. This response is sensible at the methodological level, but it does not remove the conceptual problem.

Simplicity is a criterion of preference, not a logical consequence of the data. Two observations matter here:

1. there are several ways to define simplicity: polynomial degree, description length, naturalness of the pattern, cultural familiarity, or computational ease;
2. a test that does not explicitly state its simplicity criterion asks the respondent to infer that criterion tacitly.

It follows that the "correct" answer is not merely a mathematical answer; it is also a *conventional* answer.

Remark 1. The problem is not limited to polynomials. The more general form

$$F(x) = P(x) + W(x)H(x)$$

shows that if H is arbitrary, one obtains an even larger space of compatible continuations. The polynomial restriction is used here only because it permits an elementary and fully explicit proof.

5. Computational Illustration in the Repository

5.1. What the code actually does

To make underdetermination visually intuitive, the repository includes a small computational pipeline:

- the file `lagrange.cu` generates many instances of the family $P_k(x) = P(x) + kW(x)$ on the GPU;
- the file `output.csv` stores the minimum and maximum envelope of the sampled values over a grid of points;
- the file `plot.py` produces the final figure `lagrange_plot.png`.

More specifically, the CUDA implementation:

1. reads a finite sequence of values;
2. associates the terms with the indices $x = 0, 1, \ldots, n-1$;
3. builds the Lagrange interpolating polynomial;
4. samples 1024 values of k uniformly at random in $[-5, 5]$;
5. evaluates the corresponding polynomials at 512 points of the interval $[-0.5,\ n + 0.5]$;
6. saves, for each grid point, the minimum and maximum values observed across all sampled curves.

5.2. How to read the figure

Figure 1 does not prove the theorem – the theorem is already proved formally – but it helps the reader see what the theorem means.

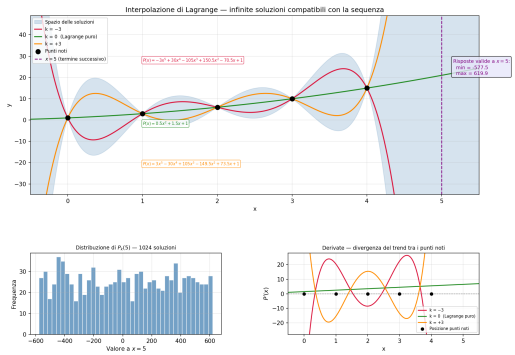

Figure 1: Visualization of the family of interpolants $P_k(x) = P(x) + kW(x)$ produced by the repository code. The blue region shows the envelope of the sampled solutions; the three colored curves show individual examples; the histogram illustrates the distribution of values obtainable at the next point; the derivative panel shows how sharply the behavior between observed points may diverge even when all curves coincide exactly on those points.

The visual message of the figure is simple:

- all curves pass through the same known points;
- between one observed point and the next, and especially beyond the last point, they may nevertheless diverge substantially;

- the value at the next point is not fixed by the data but depends on the additional parameter k.

The histogram of the value at $x = n$ is nearly uniform because, once the data are fixed, the value

$$P_k(n) = P(n) + kW(n)$$

depends linearly on k. Sampling k uniformly therefore also samples the predicted value at the next point uniformly, up to the finite effects of numerical sampling.

6. What the g Factor Is

6.1. Historical origin of the idea

The g factor takes its name from the word *general*. The idea goes back to Spearman, who observed a robust empirical pattern: scores obtained by different people on different cognitive tasks tend to be positively correlated with one another [1]. Intuitively:

> someone who performs relatively well on a verbal reasoning task tends, on average, also to perform relatively well on numerical, spatial, or memory tasks, even though clear differences in profile obviously remain.

This regularity is often called the *positive manifold*. It is one of the most replicated findings in the psychometrics of intelligence [9, 12].

6.2. An intuitive definition

The g factor is not a question, not a single task, and not a number read directly from a psychological instrument. It is a *latent factor*: a variable not directly observed, introduced in order to explain the portion of variance that many cognitive tests appear to share.

Stated even more simply:

> g is the name given to the common part that emerges when many different cognitive tests "move together" in the data.

This point is essential. The g factor is a statistically constructed quantity that receives a theoretical interpretation; it is not an object observed directly.

6.3. What g is *not*

To avoid confusion, a few distinctions are worth stating explicitly.

- **g is not identical to a raw IQ score.** An IQ score is usually a standardized index derived from a specific battery; g is the latent common factor estimated from the relations among tasks.
- **g does not exhaust every theory of intelligence.** There are hierarchical models, multifactor models, and alternative explanations of the positive manifold [3, 4, 6, 7, 9, 12].
- **g does not automatically imply a single simple cause.** Some authors interpret it as the index of a broad general capacity; others interpret it as an emergent product of partially overlapping cognitive processes [9, 12].

6.4. The relation between g, broad abilities, and specific abilities

Contemporary psychometric literature often describes intelligence in hierarchical terms. In models inspired by the work of Cattell, Horn, and Carroll, one may distinguish at least three conceptual levels [3, 4]:

1. **specific abilities**: performance in narrowly defined tasks;
2. **broad abilities**: for example fluid intelligence (G_f), crystallized intelligence (G_c), memory, or processing speed;
3. **the general factor**: the common component that stands above multiple broad abilities.

In this language, sequence items are usually interpreted as tasks that load especially on abstract reasoning and rule induction, often close to what is called fluid intelligence. But the fact that an item loads on g does not mean that its semantic content is irrelevant. This is exactly where the question of construct validity arises.

7. How the g Factor Is Calculated

7.1. The general idea

Saying that one "calculates g" can be misleading. In reality, g is not measured in the same way that temperature is measured with a thermometer. More precisely:

1. scores are collected from many people across many cognitive tests;
2. the way those scores correlate is examined;
3. a statistical model is estimated in order to summarize the shared part of the observed performance.

The final result is an estimate of a common latent factor. That estimate always depends, at least in part, on the chosen battery, the sample under study, and the statistical method that is used.

7.2. The stages of the calculation, explained without unnecessary jargon

7.3. Step 1: collecting the scores

Suppose we have N people and p cognitive tests. Let x_{ij} denote the score obtained by person i on test j. The raw data therefore form an $N \times p$ matrix.

Intuitively, each row is a person and each column is a task (vocabulary, analogies, numerical series, mental rotation, memory, and so on).

7.4. Step 2: standardization

Different tests may use very different scales. One task may range from 0 to 20, another from 200 to 800. To make them comparable, the scores are often transformed into standardized scores:

$$z_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j}, \tag{5}$$

where $\bar{x}_j$ is the mean score on test j and s_j its standard deviation.

Interpretation: $z_{ij} = 0$ means "average"; $z_{ij} = 1$ means "one standard deviation above the mean"; $z_{ij} = -1$ means "one standard deviation below the mean".

7.5. Step 3: the correlation matrix

Once the scores have been standardized, one computes the correlation matrix

$$R = (r_{jk})_{j,k=1}^{p},$$

where each coefficient r_{jk} measures the degree of association between tests j and k.

If many coefficients are positive, one observes the positive manifold that historically motivated the introduction of g [1].

7.6. Step 4: the factor model

In a simplified one-factor version, the standardized score of person i on test j is written as

$$z_{ij} = \lambda_j g_i + \epsilon_{ij}. \tag{6}$$

Here:

- g_i is the latent position of person i on the general factor;
- λ_j is the loading of test j on g;
- ϵ_{ij} contains what remains: test-specific variance, measurement error, and variability not explained by the common factor.

If λ_j is large in absolute value, that test depends strongly on the general factor. If it is small, the test reflects less g and more task-specific components.

In matrix form, the common-factor model can be summarized as

$$R \approx \Lambda\Lambda^\top + \Psi, \tag{7}$$

where Λ contains the loadings and Ψ the non-shared part.

7.7. Step 5: extracting the factor

Operationally, the model parameters must be estimated. Several procedures are available:

- exploratory factor analysis;
- confirmatory factor models;
- hierarchical or bifactor models.

All of these approaches try, in different ways, to summarize the covariation among tests. The key point for the non-specialist reader is this:

> the value of g does not simply fall out of the data automatically; it emerges from a statistical model chosen by the researcher.

7.8. Step 6: computing the individual factor score

Once the loadings have been estimated, one may assign to each person an estimate of their score on g, often denoted by $\hat{g}_i$. In schematic form:

$$\hat{g}_i = w_1 z_{i1} + w_2 z_{i2} + \cdots + w_p z_{ip}, \tag{8}$$

Step	Operation	Intuitive meaning
1	Administer a battery of different tests	One needs performance across non-identical tasks in order to see what they share.
2	Standardize the scores	Tests use different scales; standardization makes them comparable.
3	Build the correlation matrix	One measures how strongly results on different tests rise and fall together.
4	Extract the dominant common factor	One estimates the latent component that explains the largest share of shared covariation.
5	Estimate the *loadings*	For each test, one assesses how strongly it depends on the general factor.
6	Compute individual *factor scores*	For each person, one produces an estimate of their position on the general factor, given that battery and that method.

Table 1: Intuitive outline of the estimation of the g factor.

where the weights $w_1, \ldots, w_p$ depend on the scoring method used (regression scoring, Bartlett scoring, and so on).

This point is often overlooked in popular discussions:

> an individual g score is a *model-based estimate*, not a directly observed quantity.

7.9. A purely illustrative mini-example

Suppose a battery contains four tests: vocabulary, numerical series, mental rotation, and working memory. After factor analysis, one might obtain hypothetical loadings like those shown in Table 2.

A person who obtains high standardized scores across several tasks will normally receive a higher estimate of $\hat{g}$. But that estimate depends on the weights, the loadings, and the battery. Changing the test or the method may yield a non-identical estimate.

7.10. Important distinction: first principal component versus g

Remark 2. In practice one sometimes hears that g is simply the first principal component. That formulation is convenient but imprecise. PCA summarizes the total variance of the observed scores; common factor analysis instead tries to model the shared variance among tasks. The two outcomes may resemble each other, but they are not conceptually the same thing.

8. What Follows for Construct Validity

8.1. First clarification: the critique does not say that tests "measure nothing"

This distinction is crucial. From the fact that a sequence item is underdetermined, it does *not* follow that every intelligence test is useless or lacks predictive power. Indeed, the existence of the positive manifold and the ability of some batteries to predict educational outcomes are well documented [5, 10].

The critique developed here is narrower:

> even when a test is useful, the question remains whether it measures exactly the construct it claims to measure and whether each of its items is interpreted in the way it is ordinarily presented.

8.2. What construct validity is

Definition 2 (Construct validity)**.** A test has construct validity if the interpretation of its scores as measures of a given theoretical construct is justified by evidence and by a coherent theory of the process that produces the responses [2, 8].

In intuitive terms, construct validity answers the question:

> "when I observe this score, do I genuinely have good reasons to say that I am measuring the psychological property the test claims to measure?"

Good correlations with other tests or with certain external outcomes are important, but they do not

Test	Loading on g	Estimated shared variance (λ^2)
Vocabulary	0.72	0.52
Numerical series	0.78	0.61
Mental rotation	0.69	0.48
Working memory	0.64	0.41

Table 2: Illustrative example of factor loadings. The values are shown only for didactic purposes.

by themselves exhaust the problem of what the score means.

8.3. Where the underdetermination argument applies

In the case of numerical completion items, the point is not that such items are always bad items. The point is that their interpretation requires one more level of precision.

If the observed sequence does not by itself determine a unique continuation, then full credit awarded to one specific answer rewards at least one of the following abilities:

1. recognizing the pattern the test designer *intended*;
2. applying an implicit criterion of simplicity or familiarity;
3. accepting the framing of the question, namely the idea that there must be one expected continuation;
4. working quickly within shared but unstated conventions.

All of these abilities may be real and useful. But they do not automatically coincide with "general intelligence" in the broadest philosophical sense of the term.

8.4. The role of the g factor in the standard defense

The standard psychometric defense often runs as follows:

1. scores on many cognitive items correlate positively;
2. a general factor g is extracted from that structure;
3. an item that loads on g is treated as a good indicator of general intelligence.

This line of reasoning is not without force. But it does not settle the whole issue. An item may load on g and still reward more specific competences than the term "intelligence" suggests.

8.5. The circularity that should be avoided

Precision matters here. It would be incorrect to say that the entire psychometrics of g is an empty tautology. It is not. There are real data, replicated correlations, and relations with external outcomes. Nonetheless, there is a *partial circularity* that deserves attention:

- items are selected in part because they show good correlations with other tests in the battery;
- the factor g is estimated from batteries constructed in that way;
- the fact that an item loads on g says a great deal about its coherence with the battery, but by itself it does not fully answer the question of what cognitive content the item actually rewards.

In other words, *a high loading on g* and *clarity of construct* are not synonyms.

8.6. What a sequence item plausibly measures

In light of the preceding discussion, a more careful and more precise formulation might be the following:

> sequence items measure, at least in part, the ability to infer a conventional rule judged appropriate by the testing context, under time constraints and with incomplete information.

This formulation is less sweeping than "it measures intelligence in general", but it is probably closer to what the observed behavior actually supports.

9. Comparison with Broader Perspectives on Intelligence

Critiques of reducing intelligence to a single index are not new. Sternberg argued, in different forms, that traditional tests capture only part of the abilities that matter and that one must also consider creative, practical, and meta-cognitive components [6,7]. Stanovich argued that intelligence tests, even when useful, leave out dispositions that are central to rational judgment, evidence evaluation, and sound everyday reasoning [11].

This literature does not by itself prove the mathematical thesis of the present paper, but it situates it within a broader framework: a test may be empirically useful and still incomplete with respect to what many people ordinarily mean by the word "intelligence".

Sequence items are especially instructive because they display, in concentrated form, the gap between:

- a well-defined performance within a testing convention;
- the more ambitious claim of measuring a general capacity for reasoning.

10. Limits of the Argument

A rigorous paper should state not only what it shows, but also what it does not show. The main limits are four.

1. **Not all test items are equally ambiguous.** Some problems include contextual clues or implicit constraints strong enough to reduce the space of reasonable interpretations.
2. **Mathematical underdetermination does not cancel practical usefulness.** In many real settings it may be entirely useful to test whether a person can quickly recognize simple, conventional patterns.
3. **A single item is not the same as the whole battery.** Psychometric batteries aggregate many tasks precisely in order to reduce random error and increase overall reliability.
4. **The argument primarily targets interpretation of the construct.** Its main target is not the mere existence of correlations or predictive validity, but the conceptual leap from those facts to the claim that one numerical answer is the only logically correct one and that recognizing it straightforwardly coincides with general intelligence.

11. Conclusions

The central thesis of the article can be summarized in one sentence:

> a finite sequence, taken by itself, does not uniquely determine the next term; to obtain a unique answer, one must introduce additional constraints or conventions.

The Lagrange interpolation theorem and the construction of the family $P(x) + W(x)H(x)$ make it possible to establish this point in a simple, formal, and highly general way. From this it does not follow that intelligence tests are worthless; it does follow that the interpretation of their sequence items must be more sober and more precise.

When correctly understood, the g factor is a latent factor extracted from the correlational structure among cognitive tests. It is a powerful and historically central statistical tool in psychometrics. But the fact that an item loads on g is not, by itself, enough to settle every question about what that item means. Construct validity also requires an analysis of the cognitive process the item rewards.

In the case of numerical completion items, the most defensible conclusion is therefore the following: they certainly measure something, but that something is better described as the ability to infer and apply a conventionally expected rule within a testing context than as direct and unproblematic access to intelligence in a fully general sense.

A more transparent design could improve this situation in at least two ways:

1. by explicitly stating the admissible class of rules;
2. by rewarding not only the final answer but also the quality of the justification and the ability to recognize when a problem is underconstrained.

A. Essential Glossary

Interpolation The construction of a function that passes exactly through a given set of data points.

Underdetermination
A situation in which the available data are insufficient to select a unique solution without further assumptions.

Parsimony A preference for simpler explanations or models, all else being equal.

Correlation A measure of the extent to which two variables tend to rise or fall together.

Latent variable
A theoretical variable not directly observed, introduced to explain regularities in observed data.

Factor loading
A coefficient indicating how strongly a test is associated with a latent factor.

Factor score A numerical estimate, for an individual, of that person's position on a latent factor.

Construct validity
The theoretical and empirical justification for interpreting a score as a measure of a given psychological property.

Technical Appendix on the Repository

The repository includes a computational demonstration consistent with the mathematical thesis of the paper:

- `lagrange.cu` computes the family of perturbed polynomials in parallel;
- `output.csv` contains the numerical envelope of the solutions;
- `plot.py` constructs the final figure;
- `lagrange_plot.png` visualizes the result.

This empirical component does not replace the theoretical argument, but it has important pedagogical value: it makes visible, at a glance, how wide the family of solutions compatible with the same observed points can be.

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